Brian Kang

bsg.kang@mail.utoronto.ca | LinkedIn | Portfolio | +1 647 901 9338

Education

Robotics | Engineering Science - University of Toronto — Sep 2022 - Apr 2027

- **3.71 GPA** - *Dean's Honour List*

Experience

Trackside Engineering Placement - Red Bull Powertrains F1 — July 2025 - Present

Product Validation & Data Science Engineer

- Development of a lumped-parameter **thermal model of the 2026 MGU-K**, deployed **on the race ECU** achieving **±1 °C temperature prediction** live across all 4 cars to satisfy **FIA regulatory requirements**.
- Performed **competitor analysis** of energy deployment strategies, **reverse-engineering battery usage profiles** from onboard audio via **FFT and spectrogram analysis** to inform lap simulation and race strategy.
- Built a **genetic algorithm optimizer** to determine **laptime-optimal power unit allocation** across a full race season, updating dynamically as races occur to continuously re-optimize component usage.
- Authored **200+ lap dyno test specifications** for **power unit dyno running**, defining circuit selections, lap profiles and coolant configurations to generate datasets required for modelling.
- Developed a **Streamlit telemetry visualization platform**, deployed on the company **Kubernetes cluster**, enabling engineers across the factory to interactively generate custom plots from dyno data.

Power Unit Engineer

- Operated **Formula 1 power units on dynamometer test cells** to conduct durability testing and new engine pass-off programs, **monitoring real-time telemetry across engine, turbo, motor, and battery** systems to ensure parameters remain within performance and reliability envelopes.
- Diagnosed faults and anomalies during dyno running, **investigating issues across all subsystems** to trace root causes, coordinating with designers to determine solutions, and **developing diagnostic methodologies** for identifying and responding to recurring issues.
- Owned the **power unit operational limits document** defining safe operating boundaries across all systems, used as the reference by factory and trackside teams at every race weekend.

Personal Projects

6-DOF Robotic Manipulator - End-to-End Design & Build — July 2026 - Present

- Designed and built a **6 DOF robot arm** with a **80 cm reach**, integrating six stepper motors, TMC2209 /TMC5160 drivers, magnetic and mechanical limit sensing, and a **STM32-based motion control system**.
- Designed the complete **electromechanical system**, including custom mechanical arm linkages, **torque reduction**, power distribution, **motor driver circuitry**, and **embedded control hardware**.
- Currently developing **forward/inverse kinematics** and **trajectory generation** for coordinated multi-axis motion, with planned integration of ROS and MATLAB/Simulink for higher-level control and simulation.

Machine Learning Soccer Analytics Tracker — July 2024 - Aug 2024

- Developed a computer vision model using **PyTorch, OpenCV, and YOLOv5** to perform object detection and classification of players and ball, achieving **95% detection accuracy**
- Implemented tracking and analysis algorithms to compute **team possession metrics and player speed**

Skills

Programming: Python, C/C++, Java, MATLAB, Simulink
ML & Robotics: PyTorch, TensorFlow, ROS
Embedded Systems: STM32, ESP32, RISC-V Assembly, FPGA, SystemVerilog, Quartus, ModelSim
CAD: OnShape, KiCad, Fusion360, Solidworks, AutoCAD, SPICE
Hardware: 3D Printing, Laser Cutting, Oscilloscope, Soldering, Machining
F1: ATLAS, SystemMonitor, Telemetry Analysis